Physical Communication 75 (2026) 103029

Contents lists available at ScienceDirect

Physical Communication

journal homepage: www.elsevier.com/locate/phycom

Full length article

A novel beam training method for near-field wideband extremely large antenna arrays

Deyang Zhou, Ke Huang, Jingyi Guan, Yang Wang, Jianhe Du *

The School of Information and Communication Engineering, Communication University of China, 100024, Beijing, China

ARTICLE INFO

Keywords:
ELAA
Wideband
Beam training
Far-field
Near-field

ABSTRACT

Extremely large antenna arrays (ELAA) is an important technology for future sixth-generation (6G) wireless networks and enables significantly high data transmission rates. However, existing beam training methods either require high overhead or perform poorly under limited training resources due to the joint estimation of angle and distance in the near-field. To address this problem, we design a novel three-stage beam training method for near-field wideband ELAA systems. Specifically, the base station (BS) first employs wide beams designed based on beam split analysis and sub-array partitioning for initial angle estimation to narrow the angle search space. Refined angle estimation of the user equipment (UE) is then achieved using symmetric narrow beams in the second stage. Finally, distance estimation is performed by leveraging the refined angle estimates to identify the optimal beam in the near-field. Simulation results show that our method achieves a near-optimal achievable rate with low training overhead. In addition, the proposed method exhibits smaller angle estimation errors compared with the existing beam training schemes.

1. Introduction

Extremely large antenna arrays (ELAA) has been regarded as one of the core technologies for future sixth-generation (6G) networks [1,2]. Owing to its extremely large antenna arrays, ELAA can significantly improve spatial multiplexing gain and beamforming gain, thus largely enhancing the system data transmission rate [3]. Moreover, such capabilities also provide greater beamforming flexibility and strengthen physical-layer security, making it a key enabler to meet the high-capacity and high-rate demands of 6G communications [4]. In particular, at millimeter-wave (mmWave) and terahertz (THz) frequency bands, ELAA can provide high beamforming gains to compensate for the severe path loss, further extending the application of high-frequency networks [5].

In ELAA systems, the accurate channel state information (CSI) is essential for effective beamforming [6,7]. CSI is usually obtained through the channel estimation and the beam training [8]. For conventional large-scale antenna array systems, channel estimation methods can achieve satisfactory rate performance with relatively low pilot overhead [9]. However, since the dimension of the channel grows proportionally with the number of antennas, the channel dimension in ELAA systems is typically very high [10]. Consequently, conventional channel estimation schemes lead to unacceptable pilot overhead. To avoid acquiring the full high-dimensional CSI, various beam training schemes have been proposed. In [11] and [12], the authors focused on reconfigurable intelligent surface-assisted scenarios, while the works [13–15] addressed mmWave large-scale antenna arrays systems, with both sets of works proposing hierarchical codebook-based and low-complexity frameworks. Compared with conventional channel estimation, these beam training schemes only obtain partial CSI, but can still achieve the near-optimal rate performance while significantly reducing pilot overhead.

In traditional far-field beam training, the transmitter sequentially sends the codewords from a predefined codebook to scan the target space, and the physical direction of the user equipment (UE) is determined by the codeword corresponding to the maximum received power. However, unlike the far-field propagation in the fifth-generation systems, the near-field propagation becomes a critical factor in ELAA deployments, especially at high frequencies [16]. In such scenarios, the electromagnetic waves can no longer be approximated as planar waves but must be modeled more accurately as spherical waves [17]. Different from far-field planar waves, near-field spherical waves contain both the angular and distance information of the target, which makes the channel characteristics more complicated [18]. A direct consequence is that conventional beam training methods based on far-field assumptions are no longer effective in the near-field [19], since they only rely on the angle

* Corresponding author.
E-mail addresses: afobbi@cuc.edu.cn (D. Zhou), huangke@mails.cuc.edu.cn (K. Huang), 202420081000121@mails.cuc.edu.cn (J. Guan), wangyang@mails.cuc.edu.cn (Y. Wang), dujianhe1@gmail.com (J. Du).

https://doi.org/10.1016/j.phycom.2026.103029
Received 25 October 2025; Received in revised form 4 January 2026; Accepted 29 January 2026
Available online 31 January 2026
1874-4907/

search to determine the best beam direction and cannot distinguish the distance between the UE and the base station (BS) [20]. Furthermore, since both the angle and distance need to be jointly estimated, the pilot overhead required for UE position acquisition increases significantly [21,22]. Therefore, it is essential to design low-overhead and efficient near-field beam training schemes for ELAA systems.

Currently, the beam training in the far-field has been widely studied. Specifically, in [23] and [24], the authors introduced the classical discrete Fourier transform (DFT) codebook and the exhaustive beam training method, where the codebook was designed to cover the entire target space, and the beamforming vectors corresponding to the predefined directions were sequentially transmitted to the UE during the training phase. After training, the target direction was determined by the codeword achieving the maximum received power. In [25], a statistics-assisted beam training scheme was proposed to improve the spectral efficiency, where beams are scanned based on their utilization frequency and the process stops once the beamforming gain exceeds a threshold. The authors in [26] proposed a hierarchical training scheme for mmWave reconfigurable intelligent surface systems, which employs a hierarchical codebook to achieve more efficient beam alignment with only slight performance degradation. Although the literature [23–26] has achieved the beam training in the far-field, the significant overhead brought by these schemes cannot be ignored, especially in ELAA systems. To address this issue, several improved schemes have been proposed, which can generally be divided into two categories, namely multi-beam search schemes and angle calculation schemes. For the first category, in [27], a multi-beam training method was proposed for reconfigurable intelligent surface assisted multi-user communication, where the reflecting elements are divided into multiple sub-arrays to simultaneously form multiple beams over time. Moreover, the work [28] developed an efficient hybrid far/near-field beam training scheme that exploits the approximate orthogonality of near-field steering vectors to construct a multidirectional training sequence for the simultaneous estimation of angle and distance. A covert multi-user beam training strategy was introduced in [29], where a multi-finger beam codebook and a friendly jammer are jointly utilized to improve the covert throughput and alignment probability. In [30], an efficient estimation scheme based on sparse sensing matrix was devised to identify the optimal alignment with minimal training overhead. For the second category, the authors in [31] and [32] calculated the angles of departure and arrival by utilizing adjacent auxiliary beams and deriving ratio measures. To ensure flexible and efficient coverage in mmWave communications, a wide beamforming scheme employing sub-array division was proposed in [33] to design wide beams and cover the target area. Moreover, the authors in [34] further extended the ideas in [31–33] by leveraging the power distribution pattern of wide beams to develop an analytical beam training framework. This framework achieved efficient angle estimation and near-optimal achievable rate with reduced training overhead.

The massive antenna arrays in wideband ELAA systems necessitate a practical beam training scheme based on the near-field spherical wave model. Unlike the far-field assumption, the near-field spherical wave model depends on both the angle and the distance. Therefore, directly applying the far-field beam training method to the near-field channel will lead to a significant performance degradation. Furthermore, the near-field beam training is very sensitive to angle estimation, with a slight angle mismatch leading to very serious performance loss [35]. Several near-field beam training schemes have been proposed to address this issue. Similar to the far-field exhaustive beam training, the authors in [36] and [37] introduced a near-field exhaustive beam training method that performs a joint search over angles and distances using a pre-designed polar-domain codebook to determine the angle and distance of the target.

Although the above works solved the problem of angle mismatch in the near-field, the exhaustive training scheme suffered from excessively high pilot overhead, reaching ten times that of the far-field exhaustive method [38]. In order to reduce the overhead of the near-field beam training, the authors in [39] observed the relationship between far-field and near-field beams and proposed a near-field angle beam-based beam training scheme. This method first estimated the angle using far-field narrow beams and then obtained the distance based on the near-field angle-distance pairs, successfully reducing the training overhead. Moreover, inspired by the far-field beam training, a near-field hierarchical training scheme was proposed in [40] to identify the target location by uniformly searching multiple angles and distances. However, the method in [40] has limitations due to the distance parameter of the near-field channel model, which necessitates a non-uniform distance search to improve estimation accuracy. By combining the concepts of angle beam-based training and hierarchical beam training, the authors in [41] proposed a near-field two-stage hierarchical beam training method, which significantly reduced beam training overhead. In addition, several machine learning-based near-field beam training methods have been proposed. The authors in [42] designed a convolutional neural network-based beam training method, eliminating the need for predefined codebooks. In [43], a deep neural network-based scheme leverages supplementary codewords and the probability vector to refine the codeword selection, thereby jointly predicting the optimal angle and distance. Although these schemes achieve a certain reduction in training overhead, they either rely on a narrowband assumption or require considerable computational resources and execution time, rendering them impractical for current communication systems.

Inspired by the above ideas of far-field hierarchical beam training and near-field beam training schemes, we propose a novel beam training method that achieves high efficiency with low overhead for the wideband ELAA systems. By designing wide beam codebooks and utilizing the power distribution information between adjacent beams, the direction of the UE can be quickly obtained. Based on this finding, we propose a three-stage beam training method that is applicable to both near-field and far-field scenarios and has low training overhead. Specifically, our main contributions are summarized as follows.

- Firstly, we design a wide beam codebook with a custom number of beams in the angle domain. Specifically, by partitioning the antenna array into the largest possible sub-arrays, a set of wide beams with the desired angular coverage and sufficient array gain is generated to enable coarse estimation of the UE direction, thereby effectively reducing the training overhead required for subsequent narrow beam refinement. To ensure that the designed wide beams can fully exploit multi-subcarrier characteristics, the beamwidth and center direction of each wide beam are jointly designed by accounting for the beam split effect. In addition, we derive the phase compensation to mitigate the fluctuations between adjacent wide beams.
- Then, we propose a three-stage beam training method for near-field communications that significantly reduces the training overhead by decoupling angle and distance estimation. In the proposed framework, coarse angle alignment is first achieved by the designed wide beam codebook. To mitigate the array gain loss induced by near-field angle mismatch, an adjacent beam power-based refinement mechanism is introduced to localize the UE within the selected angle sector. Based on the refined angle search range, a symmetric narrow beam design is introduced to enable simultaneous transmission of two directional beams within a single time slot, thereby improving angle estimation accuracy and training efficiency. Subsequently, distance estimation is performed in the polar domain using a near-field codebook constructed from the candidate beams.
- Finally, we analyze the training overhead of existing beam training methods and provide simulation results to validate the effectiveness of the proposed three-stage scheme. Compared to existing schemes [22,39,40], the proposed method achieves superior performance in terms of training overhead, achievable rate, and angle estimation accuracy. Furthermore, the proposed method can still achieve a satisfactory rate even when training overhead is severely constrained, unlike existing beam training schemes which require more resources.

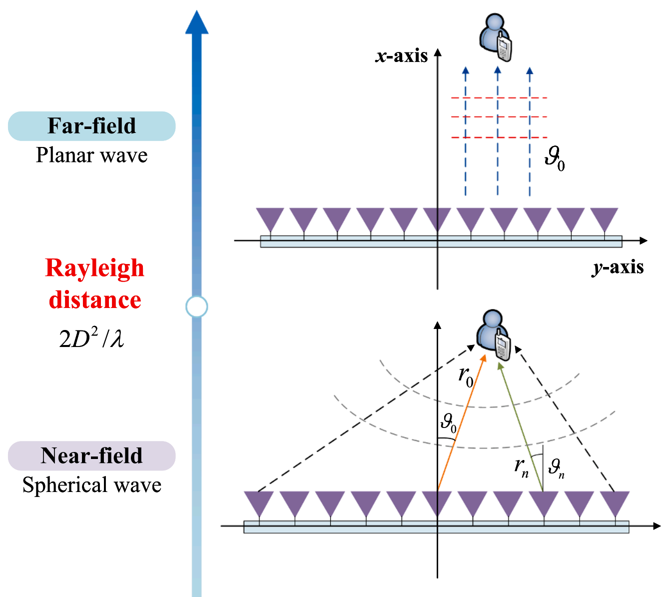


Fig. 1. The far-field and the near-field channel models.

In addition, the proposed method can also be applied to far-field channels.

The rest of this paper is organized as follow. In Section 2, the near-field and the far-field system models are introduced. In Section 3, several existing benchmark schemes for beam training are introduced. Section 4 details the wide beam codebook generation method and the proposed three-stage beam training scheme. Simulation results are shown in Section 5 and the conclusion is summarized in Section 6.

Upper-case, lower-case, and cursive letters represent matrices, vectors and tensors, respectively; $(\cdot)^T$ denotes the transpose operation; $|\cdot|$ denotes the absolute operator; $\lfloor x \rfloor$ denotes the nearest integer less than or equal to x; $\mathbf{H}[:,i]$ and $\mathbf{v}[i]$ denote the i-th column of $\mathbf{H}$ and the i-th element of $\mathbf{v}$, respectively; $\mathcal{H}[:,i,j]$ denotes the mode-1 fiber of the tensor $\mathcal{H}$ obtained by fixing the second and third indices at i and j, respectively. $\angle(\cdot)$ denotes the phase of a complex number; $\mathcal{CN}(\mu,\Sigma)$ denotes the complex Gaussian distribution with mean μ and covariance Σ.

2. System model

We consider a downlink wideband ELAA communication system where the BS is equipped with a uniform linear array (ULA) and the UE is equipped with a single antenna, respectively. The BS uses orthogonal frequency division multiplexing (OFDM) to serve the UE. We set N_t, B, M, f_c and c as the number of the antennas at the BS, the bandwidth, the number of subcarriers, the center carrier frequency and the light speed, respectively.

Due to the severe path loss caused by scattering, the main power is concentrated on the line-of-sight (LoS) path, and mmWave communication is highly dependent on the LoS path [36]. Therefore, we only focus on the LoS channel between the BS and the UE in this paper. As shown in Fig. 1, we consider two situations where the UE is in the far-field or the near-field. When the UE is in the near-field and located at $(r_0\cos\vartheta_0, r_0\sin\vartheta_0)$ with $\theta_0=\sin\vartheta_0$, the near-field channel at the m-th subcarrier between the BS and the UE $\mathbf{h}_m^{NF}\in\mathbb{C}^{N_t\times 1}$ can be defined by the Saleh-Valenzuela model as

$$\mathbf{h}_m^{NF}=\sqrt{N_t}g_m e^{-j\frac{2\pi f_m}{c}r_0}\mathbf{a}_{N_t}(\theta_0,r_0), \tag{1}$$

where g_m is the free-space path gain, $f_m=f_c+\frac{B}{M}\left(m-1-\frac{M-1}{2}\right)$, $m=1,2,\dots,M$, and $\mathbf{a}_{N_t}(\theta_0,r_0)$ is the near-field array response vector, which is expressed as

$$\mathbf{a}_{N_t}(\theta_0,r_0)=\frac{1}{\sqrt{N_t}}\left[1,e^{-j\frac{2\pi f_m}{c}(r_1-r_0)},\dots,e^{-j\frac{2\pi f_m}{c}(r_{N_t-1}-r_0)}\right]^T, \tag{2}$$

where r_n denotes the distance between the n-th BS antenna and the UE, represented as

$$\begin{aligned} r_n&=\sqrt{(r_0\cos\vartheta_0)^2+(r_0\sin\vartheta_0-nd)^2}\\ &=\sqrt{r_0^2+n^2d^2-2r_0\theta_0 nd}, \end{aligned} \tag{3}$$

where $d=\frac{\lambda_c}{2}=\frac{c}{2f_c}$ is the antenna spacing, λ_c is the central wavelength.

Due to the spherical wave property, the array response vector of the near-field differs significantly from that of far-field array. Perform a Taylor expansion on r_n with $\sqrt{1+x}\approx 1+\frac{1}{2}x-\frac{1}{8}x^2$, and we can get

$$r_n\approx r_0-nd\theta_0+n^2d^2\frac{1-\theta_0^2}{2r_0}. \tag{4}$$

When r_0 is larger than the Rayleigh distance $\frac{2D^2}{\lambda_c}$ with $D=(N_t-1)d$, the quadratic term $n^2d^2\frac{1-\theta_0^2}{2r_0}$ tends to 0. By substituting r_n into (2), we find that the near-field array response vector can be approximated to the far-field form, denoted as

$$\mathbf{b}_{N_t}(\theta_0)=\frac{1}{\sqrt{N_t}}\left[1,e^{j\frac{2\pi d f_m}{c}\theta_0},\cdots,e^{j\frac{2\pi(N_t-1)d f_m}{c}\theta_0}\right]^T. \tag{5}$$

To simplify and unify the direction, we set $\mathbf{b}_{N_t}(\varphi_m)=\frac{1}{\sqrt{N_t}}\left[1,e^{-j\pi\varphi_m},\cdots,e^{-j\pi(N_t-1)\varphi_m}\right]^T$ with $\varphi_m=\frac{2df_m}{c}\theta_0$. At the same time, the far-field LoS channel $\mathbf{h}_m^{FF}$ can be modeled as

$$\mathbf{h}_m^{FF}=\sqrt{N_t}g_m e^{-j2\pi\tau f_m}\mathbf{b}_{N_t}(\varphi_m), \tag{6}$$

where $\tau=\frac{r_0}{c}$ denotes the time delay.

Therefore, the signal received by the UE can be expressed as

$$y_m=\left(\mathbf{h}_m^X\right)^T\mathbf{v}x_m+z_m, \tag{7}$$

where $X\in\{NF,FF\}$, $\mathbf{v}\in\mathbb{C}^{N_t\times 1}$ denotes the beamforming vector at the BS, x_m denotes the transmit pilot which satisfies $|x_m|^2=1$, and z_m is the complex Gaussian noise following the distribution $\mathcal{CN}(0,\sigma^2)$ with the power σ^2.

3. Benchmark schemes

3.1. Angle-domain far-field exhaustive beam training

The conventional exhaustive beam training method in the far-field performs uniform sampling in the angle-domain and constructs a DFT codebook for beam sweeping. Specifically, a DFT codebook with size N_t can be expressed as

$$\begin{aligned}\mathbf{W}&=\left[\mathbf{b}_{N_t}(-1),\mathbf{b}_{N_t}\left(\frac{2-N_t}{N_t}\right),\cdots,\mathbf{b}_{N_t}\left(\frac{N_t-2}{N_t}\right)\right]\\ &=\begin{bmatrix}1 & 1 & \cdots & 1\\ e^{j\pi} & e^{j\frac{N_t-2}{N_t}\pi} & \cdots & e^{j\frac{2-N_t}{N_t}\pi}\\ \vdots & \vdots & \ddots & \vdots\\ e^{j(N_t-1)\pi} & e^{j(N_t-1)\frac{N_t-2}{N_t}\pi} & \cdots & e^{j(N_t-1)\frac{2-N_t}{N_t}\pi}\end{bmatrix}.\end{aligned} \tag{8}$$

During the training phase, each column in the codebook $\mathbf{W}$ is extracted as a beamforming vector and the pilots are sent to UE at different time slots. The received power at the UE can be calculated as

$$p_n=|y_n|^2=\sum_{m=1}^{M}\left|\left(\mathbf{h}_m^{FF}\right)^T\mathbf{W}[:,n]+z_m\right|^2, \tag{9}$$

where $n = 1, \dots, N_t$. Then, the best angle index i will be selected from the beamforming vector with the maximum received power, which is expressed as

$$i = \underset{1 \le i \le N_t}{\operatorname{argmax}} \; p_i. \tag{10}$$

Finally, the best transmit direction $\varphi_i = \frac{2i-N_t-2}{N_t}$ can be chosen, and the beamforming vector $\mathbf{b}_{N_t}\left(\frac{2i-N_t-2}{N_t}\right)$ at the BS can be used to transmit data.

3.2. Polar-domain near-field exhaustive beam training

Different from far-field beam training, the near-field beam training requires beam search in both the angle and the distance domains simultaneously. To address the energy diffusion effect, the near-field beam training typically employs the polar-domain codebook constructed in [19], which can be expressed as

$$\mathcal{W}[:, s, n] = \mathbf{a}_{N_t}(\theta_n, r_{n,s}), \tag{11}$$

where $s = 1, 2, \dots, S$, S denotes the number of distance samples, $\theta_n = \frac{2n-N_t}{N_t}$ denotes the angle sampling points of each codeword, $r_{n,s}$ denote the s-th distance at each θ_n, and $r_{n,s}$ is expressed as

$$r_{n,s} = \frac{1}{s}\varepsilon_\Delta\left(1 - (\theta_n)^2\right), \quad s = 1, 2, \dots, S, \tag{12}$$

where $\varepsilon_\Delta = \frac{D^2}{2\alpha_\Delta \lambda_c}$ is a constant, and it can be concluded from (12) that the near-field polar-domain codebook is non-uniformly sampled for distance.

Based on the above beam training process, the best angle and distance index is derived from the beamforming vector with the maximum received power, and the optimal codeword index for the UE is defined as

$$\left(n_{best}, s_{best}\right) = \underset{\substack{1 \le n \le N_t \\ 1 \le s \le S}}{\operatorname{argmax}} \sum_{m=1}^{M} \left| \left(\mathbf{h}_m^{NF}\right)^T \mathcal{W}[:, s, n] + z_m \right|^2. \tag{13}$$

However, the training overhead of traversing all the angle and distance indexes is $N_t S$, which is unacceptable when the number of antennas is large.

3.3. Near-field angle beam-based beam training

To reduce the high training overhead generated by the near-field exhaustive beam training scheme, a near-field training framework proposed in [39] reduces the training overhead to $N_t + S$. Specifically, the far-field beamforming vector can be used for the angle estimation in the near-field beam training. The conclusion is supported by the observation that during the far-field beam training, the actual spatial direction of a near-field UE lies near the middle of the dominant angle region with the strong beam power. Based on the above observation, the DFT codebook from Subsection III-A is used to the near-field beam training, aiming to select the angle with the maximum received power and several adjacent angles. Then, the optimal distance index for the UE is obtained based on (12) by searching across all candidate angles. Finally, the optimal angle and distance of the UE are determined via (13).

Although the above training methods can select the best beam, they cannot achieve good performance under a strict overhead constraint. Specifically, the search range for both angle and distance in the two exhaustive training methods is limited, whereas the angle beam-based beam training algorithm cannot estimate the distance of user under an overhead constraint of less than N_t, thereby causing a severe reduction in array gain. Therefore, it is necessary to design an efficient beam training method that achieves near-optimal performance with low training overhead in the near-field.

4. The proposed method

4.1. Wide beam codebook generation

The core technique of both the far-field and near-field beam training relies on narrow beams and identifying the optimal beam for data transmission. However, limited overhead prevents the training of all narrow beams, resulting in a mismatch between the beam direction and the direction of UE. Regarding this problem, we first construct a set of wide beams to determine the coarse direction of the UE, and then send narrow beams to refine the estimation. Therefore, we adopt the framework in [33] and [34], where the BS is split into multiple sub-arrays, and a conventional beamforming method is applied to each one. By employing multiple wide beams, we design a wide beam codebook that covers the entire training direction.

Specifically, we divide the antenna array at the BS into K sub-arrays, which satisfies

$$\frac{2K}{N_l} \ge \xi, \tag{14}$$

where ξ is the target beam width, and N_l is the number of antenna elements in each sub-array. Obviously, we have $KN_l \le N_t$, i.e., $K = \left\lfloor \frac{N_t}{N_l} \right\rfloor \ge \frac{N_t}{N_l} - 1$, substituting it into (14) yields

$$N_l \le \frac{\sqrt{1 + 2N_t\xi} - 1}{\xi}. \tag{15}$$

Moreover, due to $\left\lfloor \frac{N_t}{N_l} \right\rfloor \le \frac{N_t}{N_l}$, we can obtain

$$\frac{2N_t}{N_l^2} \ge \xi \iff N_l \le \sqrt{\frac{2N_t}{\xi}}. \tag{16}$$

To ensure sufficient array gain, we select the largest integer N_l that satisfies (14)-(16) as the number of antenna elements in the sub-array from the following set

$$\left[\left\lfloor \frac{\sqrt{1 + 2\xi N_t} - 1}{\xi} \right\rfloor, \dots, \left\lfloor \sqrt{\frac{2N_t}{\xi}} \right\rfloor\right]. \tag{17}$$

Then, the guiding angles of the κ-th sub-array, designed to cover the target range, can be given by

$$\nu_\kappa = \eta + \frac{2\kappa - K - 1}{N_l}, 1 \le \kappa \le K, \tag{18}$$

where η is the center direction of the wide beam. Therefore, the beamforming vectors of the BS can be written as

$$\begin{cases} \mathbf{v}\left[(\kappa-1)N_l + n\right] & = \frac{1}{\sqrt{N_t}} e^{j\partial_\kappa} e^{j\pi\left[(\kappa-1)N_l + n - 1\right]\nu_\kappa}, \\ & \quad \kappa = 1, 2, \cdots, K, \; n = 1, 2, \cdots, N_l, \\ \mathbf{v}\left[(\kappa-1)N_l + n\right] & = 0, \; \kappa > K \text{ or } n > N_l, \end{cases} \tag{19}$$

where $e^{j\partial_\kappa}$ is the phase rotation for the κ-th sub-array, and ∂_κ is phase compensation to ensure sufficient array gain and prevent serration, which satisfies $\partial_\kappa = \kappa\Delta\partial$, where $\Delta\partial$ is expressed as

$$\Delta\partial = \begin{cases} \frac{(N_l-1)\pi}{N_l} + \pi, & N_l^4 \sin^2\left(\frac{\pi}{2N_l}\right) \ge 8, \\ 2\arccos\left(\frac{\sqrt{2}}{4} N_l^2 \sin\frac{\pi}{2N_l}\right), & N_l^4 \sin^2\left(\frac{\pi}{2N_l}\right) < 8, \end{cases} \tag{20}$$

and the derivation of $\Delta\partial$ is detailed in Appendix A.

Based on the wide beam generation method described above, we can design a wide beam codebook such that a specified number of wide beams cover the target direction.

First, we denote the upper and lower bounds of the beam training angular coverage as $\vartheta_{\max}$ and $\vartheta_{\min}$, respectively. The number of wide beams is set as T. Thus, the beam width can be expressed as

$$\xi = \frac{\sin\vartheta_{\max} - \sin\vartheta_{\min}}{T}, \tag{21}$$

Then, we determine the center directions of each wide beam by

$$\eta_t = \sin\vartheta_{\min} + \left(t - \frac{1}{2}\right)\xi, \quad t = 1, 2, \ldots, T. \tag{22}$$

However, the designed wide beams inevitably suffer from power loss in wideband systems. This limitation originates from the phase shift beamformers, which can only generate frequency-independent phase shifts. As a result, the beam directions corresponding to different subcarriers deviate from each other in wideband transmission. In particular, for the subcarrier at frequency f_m, the resulting beam is steered toward $\eta_{t,m} = \frac{f_c}{f_m}\eta_t$, this phenomenon is commonly referred to as the beam split effect.

To mitigate the impact of beam split effect, a straightforward strategy is to ensure that the angle span of each beam is sufficiently wide to cover the beam deviations across all subcarriers. Then, the energy of different subcarriers remains confined within the wide beam coverage, thereby preventing severe power leakage caused by frequency-dependent angle offsets. Specifically, for the system with bandwidth B, the subcarrier frequencies lie in the range $f_m \in [f_c - B/2, f_c + B/2]$. Therefore, the magnitude of the angle deviation at the frequency f_m is given by

$$|\Delta\eta_{t,m}| = |\eta_t|\frac{|f_m - f_c|}{f_m}. \tag{23}$$

Since the frequency f_m attains its minimum at the $f_{\min} = f_c - B/2$, the max angle deviation $\Delta\eta_{\max}$ can be calculated as

$$\Delta\eta_{\max} \le |\eta_t|\frac{B/2}{f_{\min}} = |\eta_t|\frac{\beta/2}{1-\beta/2}, \tag{24}$$

where $\beta = B/f_c$ denotes the fractional bandwidth. Therefore, the half-width of the designed wide beam should be sufficiently large to cover both the half-width $\xi/2$ introduced by the angle partition and the maximum angle deviation $\Delta\eta_{\max}$ caused by the beam split effect, which can be expressed as

$$\frac{\xi_t}{2} \ge \frac{\xi}{2} + \Delta\eta_{\max}. \tag{25}$$

As a result, the required width of the wide beam can be obtained as

$$\xi_t = \xi + |\eta_t|\frac{\beta}{1-\beta/2}. \tag{26}$$

Finally, based on the angle partition defined by (21) and (26), we construct a wide beam codebook where each beam is assigned an adaptive spatial width. Specifically, according to (26), the beam width ξ_t is positively correlated with $|\eta_t|$. As a result, the designed wide beams have different widths, where beams with larger $|\eta_t|$ are intentionally widened to ensure that the beams generated by frequency-independent phase shift beamformers remain within the wide beam coverage. For example, as illustrated in Fig. 2, when $\vartheta_{\min} = -60°$, $\vartheta_{\max} = 60°$, and $T = 4$, the beam-center set is $\{-0.65, -0.22, 0.22, 0.65\}$, while the corresponding beamwidths are determined by $\xi_t = \xi + |\eta_t|\beta/(1-\beta/2)$, resulting in wider beams for larger $|\eta_t|$. The overall wide beam codebook construction procedure is summarized in Algorithm 1 .

Algorithm 1 Wide beam codebook design.

Input: The number of antennas N_t, $\vartheta_{\min}$, $\vartheta_{\max}$ and the number of the wide beams T.

1: Obtain the narrowband beamwidth ξ by (21)
2: **for** $t = 1$ **to** T **do**
3: Compute the beam direction η_t by (22)
4: Adjust the wideband beamwidth ξ_t based on the beam split effect by (23)-(26)
5: Obtain the sub-array size N_l by (14)-(17)
6: Generate the beamforming vector $\mathbf{W}_1[:,t]$ by (18)-(19)
7: Compute the phase compensation ∂_κ by (20)
8: **end for**

Output: The wide beam codebook $\mathbf{W}_1$.

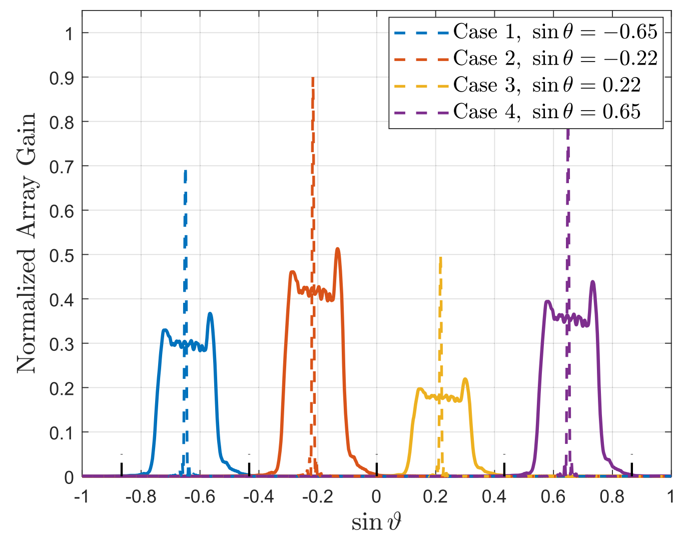


Fig. 2. The wide beam codebook pattern at $\vartheta_{\min} = -60°$, $\vartheta_{\max} = 60°$, $T = 4$ and $B = 0.25f_c$.

4.2. Proposed three-stage beam training scheme

Based on the generated wide beam codebook, we propose a novel near-field three-stage beam training scheme. Note that this method can also be applied to the far-field. In the first stage, we use the constructed wide beam codebook to send beamforming vectors and complete coarse direction estimation of the UE. The power $\bar{p}_t$ received by the UE on the t-th wide beam can be calculated as

$$\bar{p}_t = \sum_{m=1}^{M}\left|(\mathbf{h}_m^{NF})^T\mathbf{W}_1[:,t] + z_m\right|^2. \tag{27}$$

In the second stage, after determining the approximate direction, we send a series of narrow beams to refine the estimation direction of the UE within the approximate direction. However, a significant challenge arises when the training overhead is sufficient to sample all wide beams but only a limited number of narrow beams. In the case, the direction of the beams is mismatched with that of the UE, which will cause very serious power loss when the UE is in the near-field. To address this challenge, we provide a method to further refine the range of the UE using the power distribution information of adjacent wide beams.

Specifically, as shown in Fig. 3, after determining the optimal wide beam index η_b, we compare the magnitude relationship of the received power of two adjacent wide beams η_{b-1} and η_{b+1} to determine whether the current position of the UE is in the left or the right half of the wide beam. However, the array gains of different wide beams are inherently different due to the different beam widths. Therefore, it is not appropriate to directly compare the received powers among different wide beams. To address this issue, we leverage the precomputed center gain of each wide beam to mitigate the impact of array gain variations. Specifically, the center array gain of the t-th wide beam is defined as

$$\Gamma_t = \sum_{m=1}^{M}\left|\mathbf{b}_{N_t}^T(\eta_t)\,\mathbf{W}_1[:,t]\right|^2. \tag{28}$$

Then, by computing $\hat{p}_t = \bar{p}_t/\Gamma_t$, the impact of different beam widths on the array gain can be effectively mitigated. Based on the resulting power metric, if $\hat{p}_{b-1} > \hat{p}_{b+1}$, the UE is located within the left half of the optimal beam, due to the fact that the wide beam with index $b-1$ is closer to the UE and can achieve higher received power. According to the power information of adjacent beams, we can further reduce the narrow beam training range by half, thus effectively avoiding the problem of beam mismatch.

After determining the training range $[\theta_{\min}, \theta_{\max}]$ for the second stage, we send narrow beams to precisely locate the direction of the UE. The

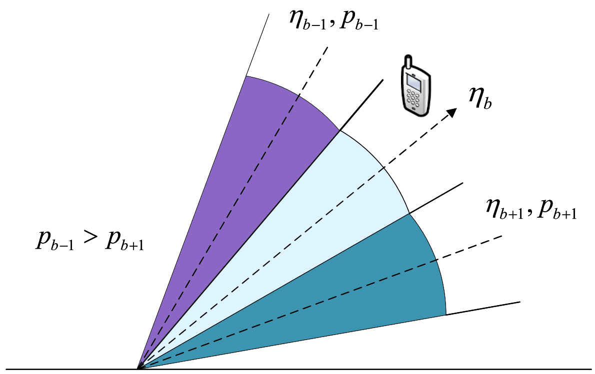


Fig. 3. The comparison of the adjacent beams. Here, $p_{b-1} > p_{b+1}$, and the UE is at the left half of the best wide beam.

directions of the narrow beams can be calculated as

$$\theta_q = \theta_{\min} + \frac{q-1}{Q-1}\left(\theta_{\max} - \theta_{\min}\right), \tag{29}$$

where $q = 1, 2, \dots, Q$, Q is the number of narrow beams that are sent to the UE. Different from the traditional far-field exhaustive method, we construct a symmetrical beam codebook by dividing the angle indexes within the estimated range into two parts and building the respective sub-codebooks. The two symmetric codebooks $\mathbf{W}_\Omega$,$\mathbf{W}_\mho$ can be respectively expressed as

$$\begin{aligned} \mathbf{W}_\Omega &= \left[\mathbf{b}_{N_t}(\theta_1), \mathbf{b}_{N_t}(\theta_2), \dots, \mathbf{b}_{N_t}(\theta_{Q/2})\right], \\ \mathbf{W}_\mho &= \left[\mathbf{b}_{N_t}(\theta_Q), \mathbf{b}_{N_t}(\theta_{Q-1}), \dots, \mathbf{b}_{N_t}(\theta_{Q/2+1})\right]. \end{aligned} \tag{30}$$

During the training phase, two symmetrical beamforming vectors are sent within one time slot to enhance the efficiency of beam training. The received powers at the UE can be respectively calculated as

$$\begin{aligned} p_q^\Omega &= \left|y_q^\Omega\right|^2 = \sum_{m=1}^{M} \left|\left(\mathbf{h}_m^{NF}\right)^T \mathbf{W}_\Omega[:, q] + z_m\right|^2, \\ p_q^\mho &= \left|y_q^\mho\right|^2 = \sum_{m=1}^{M} \left|\left(\mathbf{h}_m^{NF}\right)^T \mathbf{W}_\mho[:, q] + z_m\right|^2, \end{aligned} \tag{31}$$

where $q = 1, 2, \dots Q/2$. After traversing all candidate angles, we select the one corresponding to the maximum received power as the final estimated angle, denoted as $\hat{\theta}_\delta$, where δ is obtained from

$$\delta = \underset{1 \le q \le Q}{\operatorname{argmax}}\, \mathbf{p}, \tag{32}$$

where $\mathbf{p} = [p_1^\Omega, \dots, p_{Q/2}^\Omega, p_1^\mho, \dots, p_{Q/2}^\mho]$. Based on the above two-stage training, we can precisely estimate the angle of the UE, which is effective for the UE in both far-field and near-field channels.

Finally, in the third stage, we use the best angles estimated in the previous two stages for distance estimation, and the training overhead required in this stage is related to the number of distance sampling points. For each candidate distances under the candidate angles, the received power at UE can be calculated as

$$\tilde{p}_{\tilde{\delta},s} = \sum_{m=1}^{M} \left|\left(\mathbf{h}_m^{NF}\right)^T \mathbf{a}_{N_t}\left(\theta_{\tilde{\delta}}, r_{\tilde{\delta},s}\right) + z_m\right|^2, \tag{33}$$

where $\theta_{\tilde{\delta}} = [\theta_{\delta-1}, \theta_\delta, \theta_{\delta+1}]$, which means we take the best angle and the two adjacent angles as candidate angles. Eventually, we can get the estimated distance and direction denoted as

$$(\theta^*, r^*) = \underset{\substack{\delta-1 \le \tilde{\delta} \le \delta+1 \\ 1 \le s \le S}}{\operatorname{argmax}} \left(\tilde{p}_{\tilde{\delta},s}\right). \tag{34}$$

The illustration of the proposed three-stage training method is shown in Fig. 4 and the specific process of the method is described in Algorithm 2 . With the above training, we are able to achieve better achievable rate performance with lower training overhead compared to existing beam training schemes. Simulation results and detailed performance analysis will be presented in the next section.

Table 1
Comparison of beam training methods in terms of overhead.

Training Schemes	Total Overhead	Value
Far-field Exhaustive Training	N_t	256
Multi-directional Training [30]	$N_t/2$	128
Analytical Beam Training [34]	$N_{\text{no sp}} + 2N_{\text{sp}}$	285
Proposed Method (Far-field)	$T + Q$	16
Near-field Symmetric Training [22]	$N_t S$	3840
Near-field Angle Beam-based Training [39]	$N_t + S$	271
Near-field Hierarchical Training [40]	$\sum_{z=1}^{Z} N_x^{(z)} N_y^{(z)}$	200
Proposed Method (Near-field)	$T + Q + 3S$	77

5. Simulation results

In this section, simulation results are provided to evaluate the performance and confirm the effectiveness of the proposed three-stage beam training method. The parameters of the communication system are set to $N_t = 256$, $\alpha_\Delta = 1.44$ [19], $f_c = 40$GHz, $B = 2$GHz and the number of subcarriers is set to $M = 128$. The proposed method is compared with the analytical training method in [34], the multi-directional method in [30] and the exhaustive beam training framework in the far-field. In the near-field, the proposed method is compared with the symmetric method in [22], the angle beam-based training method in [39] and the hierarchical training method in [40].

First of all, we compare the training overhead required for different training schemes in Table 1. We set the number of angle and distance samples as $N_t = 256$ and $S = 15$. Therefore, the training overhead of the far-field exhaustive beam training scheme is 256. For multi-directional beam training, the overhead is reduced to half of the far-field exhaustive beam training by dividing two sub-arrays for beam training. By analyzing the power distribution pattern between two beams, the analytical beam training method performs angle analytical calculation for systems with beam split effects. The required training overhead depends on the sizes of the non-splitting and splitting codebooks, denoted as $N_{\text{no sp}}$ and N_{sp} respectively, and is set to 285 in our configuration. Although the near-field symmetric beam training method can slightly improve training performance, it cannot reduce the overhead. Therefore, its training overhead is the same as that of the exhaustive beam training method, given by $N_t S = 3840$. The near-field angle beam-based training method successfully reduces the overhead to $N_t + S$ by exploiting the relationship between the near-field and the far-field beams. The near-field hierarchical beam training gradually narrows the training range by constructing Z levels of sub-codebooks. In this method, each level-z sub-codebook has a distinct resolution, defined by dividing its sampling region on the $x - y$ plane into $N_x^{(z)} N_y^{(z)}$ grids, where $N_x^{(z)}$ and $N_y^{(z)}$ are the numbers of divisions along the x- and y-axes, respectively. In our simulations, we set $Z=2$ and $N_x^{(z)} = N_y^{(z)} = 10$, so the total training overhead of the hierarchical training is $\sum_z^Z N_x^{(z)} N_y^{(z)} = 200$. By carefully designing the wide beams and exploiting their power distribution, the proposed method significantly reduces training overhead compared to other schemes in both the far-field and the near-field. The numbers of wide beams and narrow beams is set to 4 and 12 in the far-field, and 16 and 16 in the near-field, respectively.

Fig. 5 shows the achievable rate versus the Signal-to-Noise Ratio (SNR), and the achievable rate is calculated as

$$\begin{aligned} \mathrm{R} &= \frac{1}{M} \sum_{m=1}^{M} \log_2\left(1 + \frac{P_t}{\sigma^2}\left|\left(\mathbf{h}_m^X\right)^T \bar{\mathbf{w}}_m\right|^2\right) \\ &= \frac{1}{M} \sum_{m=1}^{M} \log_2\left(1 + \frac{P_t N_t g_m^2}{\sigma^2}\left|\bar{\mathbf{v}}_{N_t}^T(\theta_0, r_0)\bar{\mathbf{w}}_m\right|^2\right), \end{aligned} \tag{35}$$

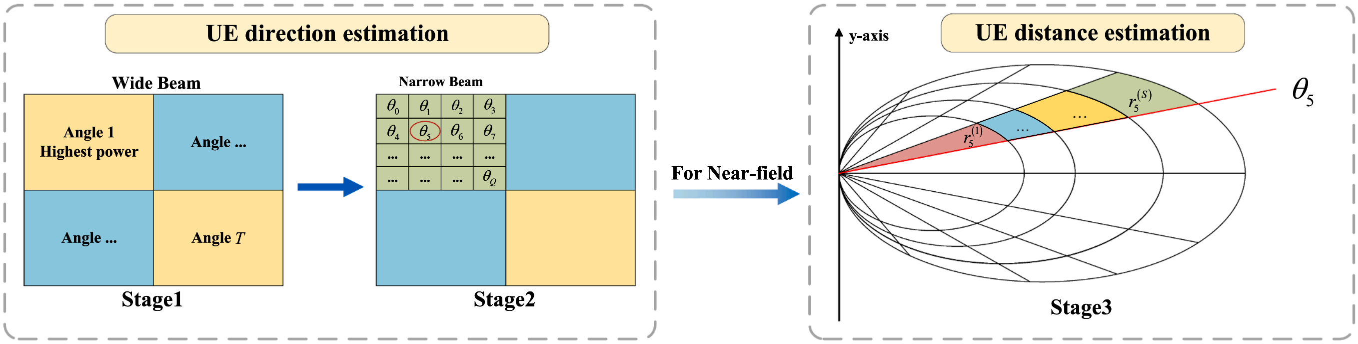


Fig. 4. Illustration of the proposed three-stage beam training method.

Algorithm 2 Proposed three-stage beam training method.

Input: The wide beam codebook $\mathbf{W}_1$, the number of narrow beams Q, and the number of wide beams T.

1: **Training stage 1:**
2: **for** $t = 1$ to T **do**
3: Compute the power $\hat{p}_t$ by (27)-(28).
4: **end for**
5: Choose the best wide beam direction η_b by the highest received power.
6: Determine the estimated range $[\theta_{\min}, \theta_{\max}]$ using adjacent wide beam power information.
7: **Training stage 2:**
8: Generate the symmetric codebooks $\mathbf{W}_\Omega$ and $\mathbf{W}_\mho$ by (30).
9: **for** $q = 1$ to $Q/2$ **do**
10: Compute the received power p_q^Ω and $p_q^\mho$ by (31).
11: Store p_q^Ω and $p_q^\mho$ into vector $\mathbf{p}_{\text{all}}$.
12: **end for**
13: Compute the best index δ by (32).
14: Obtain the estimated angle θ_δ and its adjacent angles $\theta_{\delta+1}$ and $\theta_{\delta-1}$.
15: **Training stage 3 (for near-field):**
16: Generate candidate distances $r_{\tilde{\delta},s}$ based on $\theta_{\tilde{\delta}}$ using (12).
17: **for** $s = 1$ to S **do**
18: Compute the power $\tilde{p}_{\tilde{\delta},s}$ by (33).
19: **end for**
20: Choose the optimal angle θ^* and distance r^* by (34).

Output: The estimated UE direction θ^* and distance r^*.

where SNR $= \frac{P_t N_t g_m^2}{\sigma^2}$, $\bar{\mathbf{v}} \in \{\mathbf{a}, \mathbf{b}\}$, $\bar{\mathbf{w}}_m$, P_t and σ^2 denotes the beamforming vector determined by the beam training method, the transmission power at the BS and the noise power, respectively. The SNR increases from −10dB to 20dB. The max training overhead is set to $\chi = 128$, the direction estimate range is $\left[-\frac{\pi}{3}, \frac{\pi}{3}\right]$ and the distance range is from 3m to 30m. In Fig. 5(a), we can observe that the multi-directional training method achieves higher rate than the exhaustive beam training method by dividing two sub-arrays, while the analytical beam training method is better since it can directly calculate the direction of the UE instead of selecting the optimal beam. In Fig. 5(b), the achievable rate of the angle beam-based beam training method is slightly inferior to that of the symmetric beam method. This is due to the inability of the angle beam-based method to search for distance under the constraint of overhead, whereas the symmetric beam training method improves the search efficiency and can simultaneously search both angle and distance. The hierarchical beam training scheme can achieve better rate performance than the above two methods, since its search accuracy gradually improves with the increase of the training overhead. In addition, the proposed method achieves performance comparable to other schemes in the low SNR regime, due to the fact that the wide beams generated by sub-array partitioning are more susceptible to noise. However, the existing methods all require high training overhead, which leads to the degradation of their performance when the overhead is limited. As a result, the proposed beam training scheme exhibits better performance than the existing far-field and near-field methods and achieves a near-optimal achievable rate.

To intuitively illustrate the accuracy of the beam training scheme, as shown in Fig. 6, we present the curve of angle estimation error varying with training overhead in the far-field and the near-field. The angle error is expressed as

$$\psi = \frac{|\vartheta_{\text{est}} - \vartheta_{\text{real}}|}{|\vartheta_{\max}| + |\vartheta_{\min}|}, \tag{36}$$

where the ϑ_{est} is the angle estimated by the beam training method and ϑ_{real} is the real angle. The training overhead increases from 5 to 400 and the SNR is set as 20dB. The settings for the estimation of angle and distance are consistent with those in Fig. 5. In Fig. 6(a), the angle estimation errors of the multi-directional and exhaustive beam training methods are high since they can not search the best angle when the training overhead is low. The analytical beam training method is based on the idea of angle calculation, and thus can maintain a low estimation error even with low overhead. Furthermore, we can observe from Fig. 6(b) that the angle estimation error of the symmetric training is slightly lower than the angle beam-based beam training method, owing to it can search two symmetric angles in one time slot. In addition, we can observe that the hierarchical beam training exhibits very high angle estimation errors when the training overhead is low, while its error decreases rapidly as the training overhead increases. The reason is that the hierarchical beam training cannot effectively refine the angle estimation range with a low training overhead. The proposed method exhibits comparable error to symmetric beam training under low overhead, but achieves the minimum angle estimation error when the overhead reaches approximately 50, significantly outperforming existing beam training schemes.

Then, we show the achievable rate against the distance in Fig. 7, where the distance between the BS and the UE is increased from 5m to 105m, the training overhead is set as $\chi = 128$, and the other settings are the same as those in Fig. 7. It can be observed in Fig. 7 that the proposed method achieves significantly higher achievable rate than existing beam training methods over the distance estimation range, and its performance converges at an estimated distance of approximately 15m. The angle beam-based training scheme has the lowest achievable rate since it can only search for angles. However, the high beam gain brought by the far-field narrow beam makes its achievable rate only slightly inferior to that of the symmetric beam training method. Moreover, the hierarchical beam training employs uniform sampling in the $x - y$ plane, which does not account for the distance characteristics of near-field spherical waves, resulting in fluctuations in its achievable rate over distance.

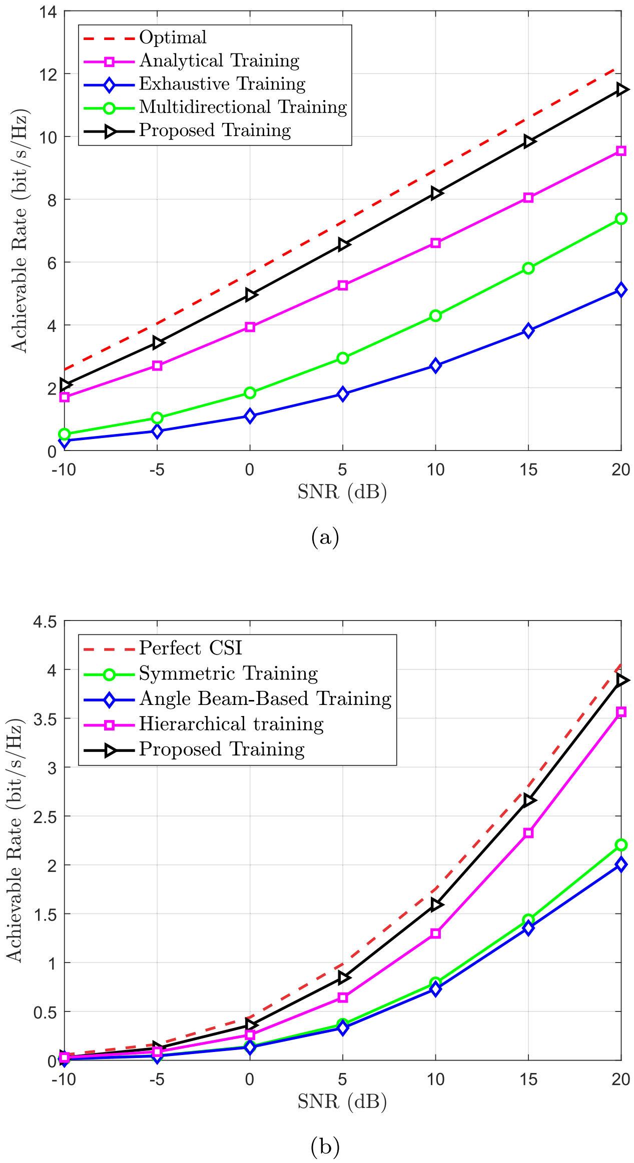


Fig. 5. Achievable rate vs. SNR. (a) Far-field, (b) Near-field.

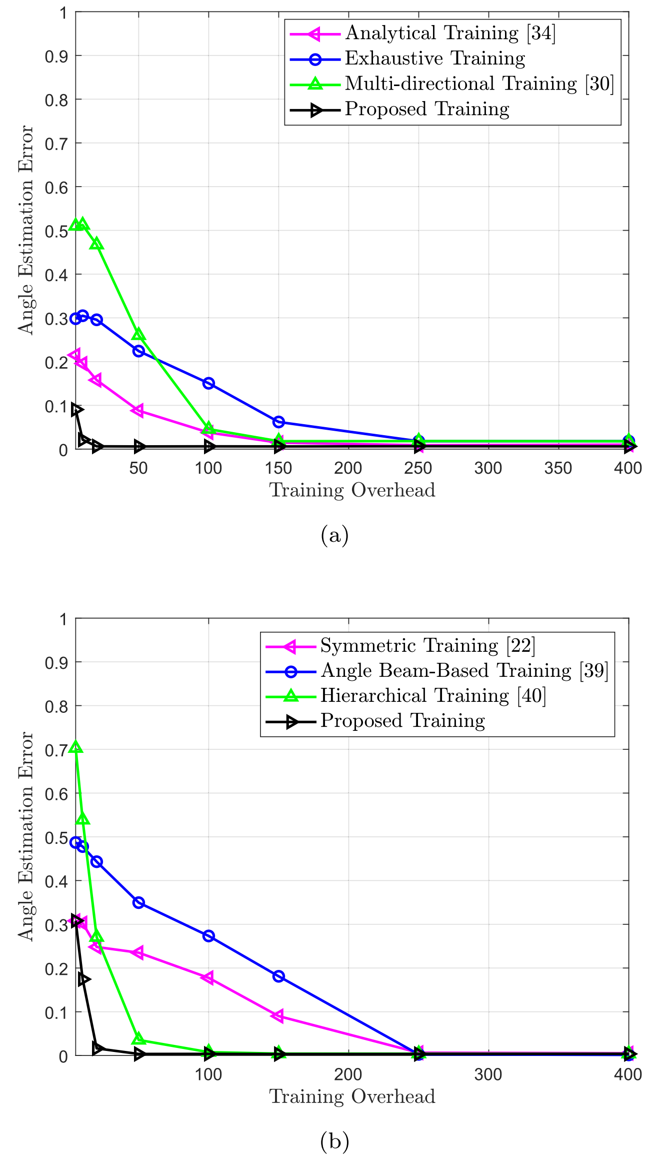


Fig. 6. Angle estimation error vs. training overhead. (a) far-field, (b) near-field.

Fig. 8 illustrates the achievable rate of different beam training schemes as the training overhead increases in order to demonstrate the superiority of the proposed method at low overhead. The beam training overhead is increased from 5 to 600, the other parameters are the same as those in Fig. 7. From Fig. 8(a), we can find that the proposed method always achieves a higher rate than the existing methods and converges quickly, even under extremely low training overhead. The achievable rate of the analytical training method fluctuates when the overhead is low, but it remains superior to the multi-directional and exhaustive beam training methods. In Fig. 8(b), we can observe that the rate performance of existing beam training methods is unsatisfactory under low training overhead, while the proposed method can achieve a near-optimal rate when the training overhead is 50. However, the achievable rate of the proposed training scheme becomes almost the same as that of the angle beam-based training method when the overhead is sufficient to search all angles and distances. This can be attributed to the fact that the angle beam-based scheme employs a higher angle sampling density, which provides a higher array gain when the training overhead is enough to search all the candidate angles and distances. Nevertheless, the proposed method can still achieve very high

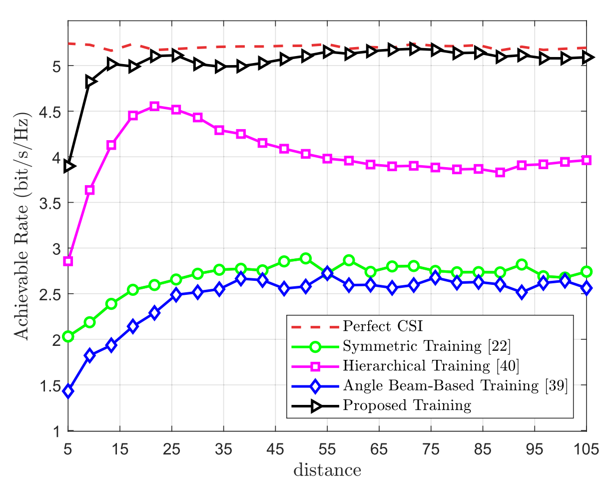
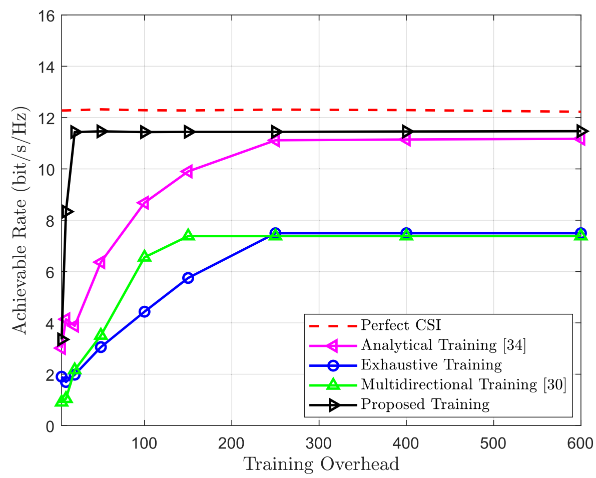
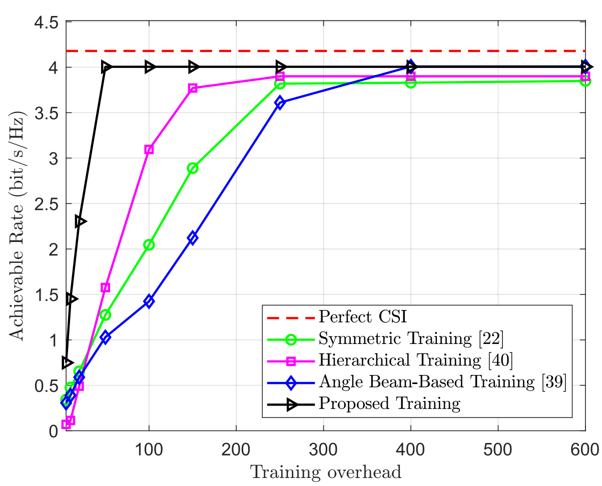


Fig. 7. Achievable rate vs. distance.

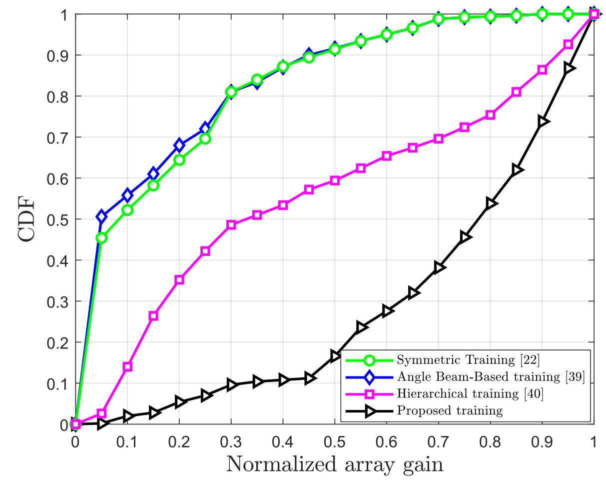


(a)

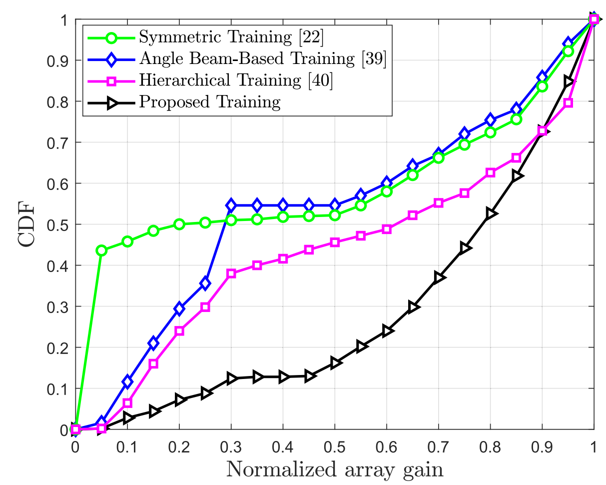


(b)

Fig. 8. Achievable rate vs. training overhead. (a) Far-field, (b) Near-field.



(a)



(b)

Fig. 9. CDF vs. normalized array gain. (a) the max training overhead $\chi = 128$, (b) The max training overhead $\chi = 1000$.

achievable rates with low overhead, a property not available in existing beam training methods.

Besides, cumulative probability functions (CDFs) of normalized array gains are shown in Fig. 9, where the SNR is set as 20dB, the angle estimate range is $\left[-\frac{\pi}{3}, \frac{\pi}{3}\right]$ and the distance range is from 3m to 30m. The max training overheads are set as 128 in Fig. 9(a) and 1000 in Fig. 9(b). It can be observed from Fig. 9 that the array gain achieved by the angle beam-based beam training and symmetric beam training methods is relatively low, whereas the hierarchical beam training method performs better. The proposed method consistently achieves the better array gain than other training methods even with sufficient training overhead. This advantage arises since the proposed method not only rapidly acquires the direction of the UE but also employs a non-uniform distance sampling strategy [19]. In contrast, the hierarchical beam training method suffers from partial array gain loss due to its use of a uniform distance sampling strategy.

Moreover, Fig. 10 illustrates the angle estimation error versus SNR, where the SNR ranges from -10 dB to 20 dB. The number of subcarriers is set to $M = 128$, 256, and 512 in Fig. 10(a)–(c), respectively. The maximum training overhead is set to $\chi = 200$, the angle search range is $\left[-\frac{\pi}{3}, \frac{\pi}{3}\right]$, and the distance range spans from 7 m to 37 m. As shown in Fig. 10(a), the proposed method exhibits relatively large angle estimation errors under low SNR regime. This can be attributed to the limited array gain provided by the wide beams employed in the first stage, which reduces the ability to combat noise. However, as illustrated in Fig. 10(b) and (c), the angle estimation error of the proposed method is significantly reduced with an increasing number of subcarriers. This improvement arises from the increased number of subcarriers under a fixed bandwidth, which leads to more frequency components being located closer to the central frequency, thereby accumulating sufficient received power to enhance robustness against noise. In contrast, the existing methods either fail to exploit wideband characteristics in beam design or require high training overhead, leading to inferior angle estimation performance.

Finally, we illustrate the impact of the bandwidth and different numbers of subcarriers on the rate performance of the proposed method in Fig. 11. The bandwidth varies from 0.1GHz to 4GHz and the SNR increases from -10dB to 10dB, while the other system settings remain the same as those in Fig. 10. It can be observed from Fig. 11 that the achievable rate of the proposed method exhibits a gradual degradation as the system bandwidth increases. This is due to the fact that the proposed method employs the phase shift beamformers to generate frequency-independent beams, which makes the beam offset across different subcarriers more pronounced in wideband systems as the bandwidth increases. Moreover, as the number of subcarriers increases, the rate per-

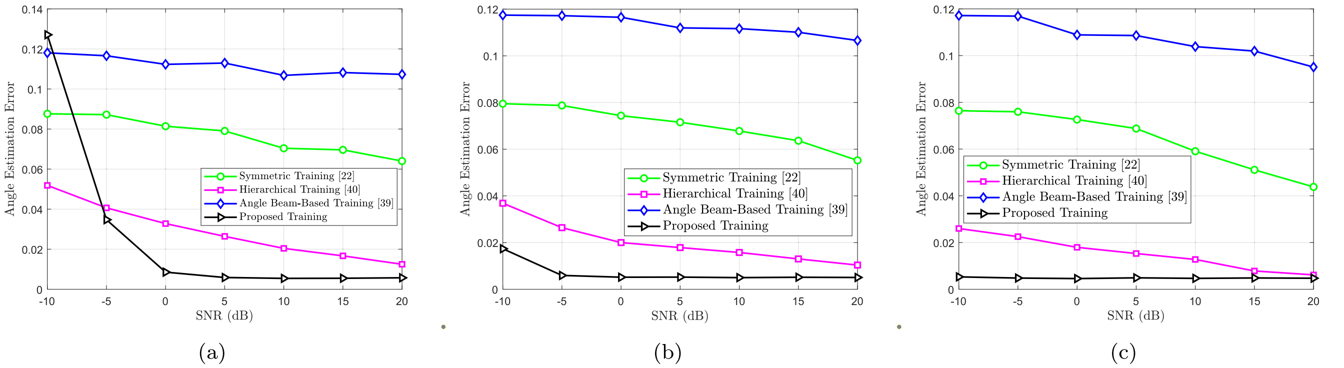


Fig. 10. Angle estimation error vs. SNR. (a) $M = 128$, (b) $M = 256$, (c) $M = 512$.

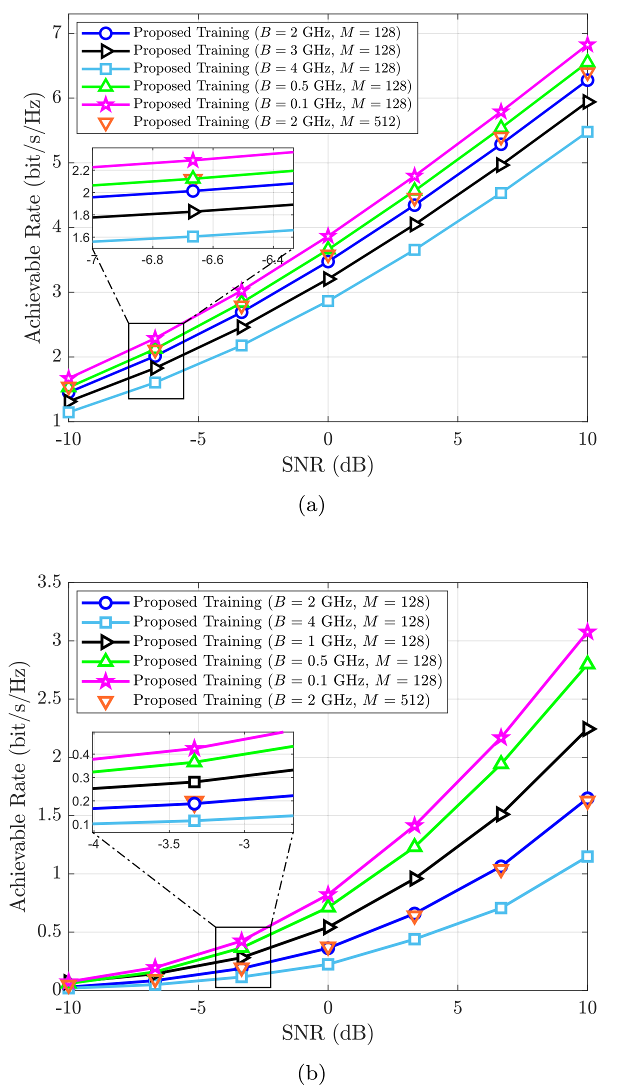


Fig. 11. Achievable rate vs. bandwidth and subcarriers. (a) far-field. (b) near-field.

formance of the proposed method is slightly improved and exhibits enhanced stability in the low SNR regime. This improvement arises from the wide beam design of the proposed method, which exploits the beam split effect and provides inherent advantages in multi-subcarrier systems.

6. Conclusion

In this paper, we tackled the beam training problems of the far-field and near-field for wideband ELAA systems. In order to reduce the high training overhead caused by the existing training method, we proposed a novel three-stage beam training method which is applicable to both near- and far-field. We first designed a wide beam codebook based on the analysis of the beam split effect and sub-array partitioning to coarsely estimate the UE direction, thereby narrowing the beam training space and reducing the overall training overhead. By leveraging the power distribution across adjacent wide beams to further refine the search range, we mitigated the problem of the array gain loss caused by the angle mismatch in the near-field. In the second stage, we constructed the symmetric DFT codebooks to further estimate the precise angle based on the range determined by the first stage. Finally, we transmitted near-field beams over the previously identified optimal angle set to determine the user's optimal angle and distance, enabling low overhead near-field beam training. Simulation results showed that the proposed beam training scheme can achieve accurate angle and distance estimation with extremely low overhead, thereby obtaining better rate performance than existing methods. Our future work will extend this study to enhance system performance by exploiting the beam split effect, while fast beam training methods based on time-delay beamformers will also be considered.

CRediT authorship contribution statement

Deyang Zhou: Writing – review & editing, Writing – original draft, Validation, Supervision; **Ke Huang:** Writing – original draft, Visualization, Methodology, Data curation; **Jingyi Guan:** Writing – review & editing, Validation, Supervision, Methodology; **Yang Wang:** Validation, Supervision, Formal analysis; **Jianhe Du:** Supervision, Resources, Project administration, Formal analysis.

Data availability

The data that has been used is confidential.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported in part by the National Natural Science Foundation of China under Grant 62471444 and in part by the Fundamental Research Funds for the Central Universities under Grant CUCZD2501 and Grant CUC25GT27.

Appendix A

Derivation of The Phase Compensation (20)

Since the beam gain of each sub-array is centered at v_κ, the beam fluctuation between two adjacent deflection angles is very severe. To solve this problem, we maximize the beam gain at $\overline{v} = \left(v_\kappa + v_{\kappa+1}\right)/2$.

The array gain of the κ-th sub-array at $\overline{v}$ can be expressed as

$$\begin{aligned}|g(\kappa)|^2 &= \left|\sum_{s=1}^{K+1}\sum_{n=1}^{N_l}\frac{1}{N_t}e^{-j\pi[(s-1)N_l+n-1]\overline{v}}e^{j\pi[(s-1)N_l+n-1]v_\kappa}e^{j\partial_s}\right|^2\\ &= \frac{1}{N_t^2}\left|\sum_{s=1}^{K+1}e^{-j\pi(s-1)(2s-2\kappa-1)}e^{j\partial_s}\sum_{n=1}^{N_l}e^{j\pi(n-1)\frac{2s-2\kappa-1}{N_l}}\right|^2.\end{aligned} \tag{37}$$

Furthermore, (37) can be approximated as

$$\begin{aligned}&\frac{1}{N_t^2}\left|\sum_{s=1}^{K+1}e^{-j\pi(s-1)(2s-2\kappa-1)}e^{j\partial_s}\sum_{n=1}^{N_l}e^{j\pi(n-1)\frac{2s-2\kappa-1}{N_l}}\right|^2\\ &\overset{(a)}{\approx}\frac{1}{N_t^2}\left|e^{j\pi(\kappa-1)}e^{j\partial_\kappa}\sum_{n=1}^{N_l}e^{-j\pi\frac{n-1}{N_l}}+e^{-j\pi\kappa}e^{j\partial_{\kappa+1}}\sum_{n=1}^{N_l}e^{j\pi\frac{n-1}{N_l}}\right|^2\\ &=\frac{1}{N_t^2N_l^2\sin^2\frac{\pi}{2N_l}}\left|e^{j\frac{(N_l-1)\pi}{2N_l}}-e^{j(\partial_{\kappa+1}-\partial_\kappa)}e^{-j\frac{(N_l-1)\pi}{2N_l}}\right|^2.\end{aligned} \tag{38}$$

Then, we prove (a), which is derived from the Lemma 1 below.

Lemma 1.

$$\left|\sum_{s=1}^{K+1}\sum_{n=1}^{N_l}e^{j\pi(n-1)\frac{2s-2\kappa-1}{N_l}}\right|^2\approx\left|\sum_{n=1}^{N_l}e^{-j\pi\frac{n-1}{N_l}}+\sum_{n=1}^{N_l}e^{j\pi\frac{n-1}{N_l}}\right|^2. \tag{39}$$

Proof. The left side of (34) can be expanded as

$$\begin{aligned}&\left|\sum_{s=1}^{K+1}\sum_{n=1}^{N_l}e^{j\pi(n-1)\frac{2s-2\kappa-1}{N_l}}\right|^2\\ &=\left|\sum_{s=1}^{K+1}e^{j\pi\frac{2\kappa-2s+1}{N_l}}\frac{\sin\left(\frac{N_l\pi}{2}\frac{2s-2\kappa-1}{N_l}\right)}{N_l\sin\left(\frac{\pi}{2}\frac{2s-2\kappa-1}{N_l}\right)}e^{j\frac{(3N_l-1)\pi}{2}\frac{2s-2\kappa-1}{N_l}}\right|^2.\end{aligned} \tag{40}$$

Given $f(s)=\left|\frac{\sin\left(\frac{N_l\pi}{2}\cdot\frac{2s-2\kappa-1}{N_l}\right)}{N_l\sin\left(\frac{\pi}{2}\cdot\frac{2s-2\kappa-1}{N_l}\right)}\right|^2$ defined for $s=1,2,\dots,K+1$, the symmetry of the function implies that its maximum value of 1 is achieved at $s=\kappa+1/2$. However, since s must be an integer, the maximum value of $f(s)$ can be attained at $s=\kappa$ and $s=\kappa+1$. If $s\neq\kappa$ and $s\neq\kappa+1$, we have

$$\frac{f(s)}{f(\hat{s})}\leq\frac{\sin^2\pi/2N_l}{\sin^2 3\pi/2N_l}. \tag{41}$$

Since N_l is relatively large, $f <= 1/9$ is insignificant and can be ignored. Therefore, the two maximum points of $f(s)$ can be retained to replace the original formula, and then the proof can be completed.

To maximize (38), we make $\sum_{n=1}^{N_l}e^{-\frac{j\pi(n-1)}{N_l}}-e^{j(\partial_{\kappa+1}-\partial_\kappa)}\sum_{n=1}^{N_l}e^{\frac{j\pi(n-1)}{N_l}}$ have the same phase for any $\kappa\in\{1,2,\cdots,K\}$, i.e.,

$$\begin{aligned}&\partial_{\kappa+1}-\partial_\kappa\\ &=\measuredangle\left(\sum_{n=1}^{N_l}e^{-\frac{j\pi(n-1)}{N_l}}\right)-\measuredangle\left(\sum_{n=1}^{N_l}e^{\frac{j\pi(n-1)}{N_l}}\right)+\pi\triangleq\Delta\partial.\end{aligned} \tag{42}$$

Therefore, we can set phase rotation for the κ-th sub-array as

$$\partial_\kappa=\kappa\Delta\partial. \tag{43}$$

Now we need to figure out the conditions that the phase compensation $\Delta\partial$ satisfies. According to (38), the total array gain at $\overline{v}$ is expressed as the sum of the gains of two adjacent wide beams, which can be expressed as

$$\begin{aligned}P&=|g(\kappa)|^2+|g(\kappa+1)|^2\\ &=\frac{1}{N_t^2N_l^2\sin^2\frac{\pi}{2N_l}}\left(\left|e^{j\frac{(N_l-1)\pi}{2N_l}}-e^{j(\partial_{\kappa+1}-\partial_\kappa)}e^{-j\frac{(N_l-1)\pi}{2N_l}}\right|^2\right.\\ &\quad\left.+\left|e^{j\frac{(N_l-1)\pi}{2N_l}}-e^{j(\partial_{\kappa+2}-\partial_{\kappa+1})}e^{-j\frac{(N_l-1)\pi}{2N_l}}\right|^2\right).\end{aligned} \tag{44}$$

Based on the similar structure of the two items in (44), we can set $P=N_l^2/N_t^2$ to ensure sufficient array gain. Meanwhile, when we set $\alpha=\frac{(N_l-1)\pi}{2N_l}$ and $\beta=\Delta\partial-\frac{(N_l-1)\pi}{2N_l}$, the first term of (44) can be simplified as

$$\begin{aligned}&\left|e^{j\frac{(N_l-1)\pi}{2N_l}}-e^{j(\partial_{\kappa+1}-\partial_\kappa)}e^{-j\frac{(N_l-1)\pi}{2N_l}}\right|^2\\ &=\left|e^\alpha-e^\beta\right|^2\\ &=4\sin^2\left(\frac{\alpha-\beta}{2}\right)\leq 4.\end{aligned} \tag{45}$$

Furthermore, substituting $P=N_l^2/N_t^2$ into (44), (45) can also be expressed as

$$\left|e^{j\frac{(N_l-1)\pi}{2N_l}}-e^{j(\partial_{\kappa+1}-\partial_\kappa)}e^{-j\frac{(N_l-1)\pi}{2N_l}}\right|^2=\frac{N_l^4\sin^2\frac{\pi}{2N_l}}{2}. \tag{46}$$

According to (45) and (46), the first term of the (44) has an upper bound and satisfies the array gain condition. Therefore, we can obtain the final expression as

$$\left|e^{j\frac{(N_l-1)\pi}{2N_l}}-e^{j(\partial_{\kappa+1}-\partial_\kappa)}e^{-j\frac{(N_l-1)\pi}{2N_l}}\right|^2=\min\left\{4,\frac{N_l^4\sin^2\frac{\pi}{2N_l}}{2}\right\}. \tag{47}$$

Due to structural similarity, the second term follows the same form as the first, thus, its derivation is omitted for brevity. Therefore, the value of phase compensation is

$$\Delta\partial=\begin{cases}\frac{(N_l-1)\pi}{N_l}+\pi, & N_l^4\sin^2\left(\frac{\pi}{2N_l}\right)\geq 8,\\ 2\arccos\left(\frac{\sqrt{2}}{4}N_l^2\sin\frac{\pi}{2N_l}\right), & N_l^4\sin^2\left(\frac{\pi}{2N_l}\right)<8.\end{cases} \tag{48}$$

References

[1] M. Cui, Z. Wu, Y. Lu, X. Wei, L. Dai, Near-field MIMO communications for 6G: fundamentals, challenges, potentials, and future directions, IEEE Commun. Mag. 61 (1) (2023) 40–46. https://doi.org/10.1109/MCOM.004.2200136.

[2] M. Giordani, M. Polese, M. Mezzavilla, S. Rangan, M. Zorzi, Toward 6G networks: use cases and technologies, IEEE Commun. Mag. 58 (3) (2020) 55–61. https://doi.org/10.1109/MCOM.001.1900411.

[3] H. Lu, Y. Zeng, Communicating with extremely large-scale array/surface: unified modeling and performance analysis, IEEE Trans. Wireless Commun. 21 (6) (2022) 4039–4053. https://doi.org/10.1109/TWC.2021.3126384.

[4] X. Li, M. Liu, S. Dang, N.C. Luong, C. Yuen, A. Nallanathan, D. Niyato, Covert communications with enhanced physical layer security in RIS-assisted cooperative networks, IEEE Trans. Wireless Commun. 24 (7) (2025) 5605–5619. https://doi.org/10.1109/TWC.2025.3548024.

[5] J. Du, Y. Cheng, L. Jin, S. Li, F. Gao, Nested tensor-based integrated sensing and communication in RIS-assisted THz MIMO systems, IEEE Trans. Signal Process. 72 (2024) 1141–1157. https://doi.org/10.1109/TSP.2024.3359323.

[6] S.H. Lim, S. Kim, B. Shim, J.W. Choi, Efficient beam training and sparse channel estimation for millimeter wave communications under mobility, IEEE Trans. Commun. 68 (10) (2020) 6583–6596. https://doi.org/10.1109/TCOMM.2020.3010024.

[7] X. Li, Z. Tian, J. Zhang, X. Xu, H. Peng, L. Li, Performance analysis of covert communication in IRS-assisted NOMA networks, Sci. Sin. Inf. 54 (6) (2024) 1502–1515. https://doi.org/10.1360/SSI-2023-0174.

[8] H. Chung, S. Kim, Location-aware beam training and multi-dimensional ANM-based channel estimation for RIS-aided mmwave systems, IEEE Trans. Wireless Commun. 23 (1) (2024) 652–666. https://doi.org/10.1109/TWC.2023.3281308.

[9] J. Du, M. Han, Y. Chen, L. Jin, F. Gao, Tensor-based joint channel estimation and symbol detection for time-varying mmwave massive MIMO systems, IEEE Trans. Signal Process. 69 (2021) 6251–6266. https://doi.org/10.1109/TSP.2021.3125607.

[10] Z. Hu, C. Chen, Y. Jin, L. Zhou, Q. Wei, Hybrid-field channel estimation for extremely large-scale massive MIMO system, IEEE Commun. Lett. 27 (1) (2023) 303–307. https://doi.org/10.1109/LCOMM.2022.3219937.

[11] Y. Xu, C. Huang, L. Wei, Z. Yang, X. Chen, Z. Zhang, C. Yuen, M. Debbah, Low-complexity beam training for multi-RIS-assisted multi-user communications, IEEE Wireless Commun. Lett. 13 (8) (2024) 2030–2034. https://doi.org/10.1109/LWC.2024.3386925.
[12] J. Wang, W. Tang, S. Jin, C.-K. Wen, X. Li, X. Hou, Hierarchical codebook-based beam training for RIS-assisted mmwave communication systems, IEEE Trans. Commun. 71 (6) (2023) 3650–3662. https://doi.org/10.1109/TCOMM.2023.3251374.
[13] K. Chen, C. Qi, Beam training based on dynamic hierarchical codebook for millimeter wave massive MIMO, IEEE Commun. Lett. 23 (1) (2019) 132–135. https://doi.org/10.1109/LCOMM.2018.2881084.
[14] W. Wu, D. Liu, X. Hou, M. Liu, Low-complexity beam training for 5G millimeter-wave massive MIMO systems, IEEE Trans. Veh. Technol. 69 (1) (2020) 361–376. https://doi.org/10.1109/TVT.2019.2926430.
[15] Z. Xiao, T. He, P. Xia, X.G. Xia, Hierarchical codebook design for beamforming training in millimeter-wave communication, IEEE Trans. Wireless Commun. 15 (5) (2016) 3380–3392. https://doi.org/10.1109/TWC.2016.2520930.
[16] Z. Wang, X. Mu, Y. Liu, Near-field integrated sensing and communications, IEEE Commun. Lett. 27 (8) (2023) 2048–2052. https://doi.org/10.1109/LCOMM.2023.3280132.
[17] Y. Chen, C. Han, E. Björnson, Can far-field beam training be deployed for cross-field beam alignment in terahertz UM-MIMO communications?, IEEE Trans. Wireless Commun. 23 (10) (2024) 14972–14987. https://doi.org/10.1109/TWC.2024.3422257.
[18] M. Cui, L. Dai, Near-field wideband beamforming for extremely large antenna arrays, IEEE Trans. Wireless Commun. 23 (10) (2024) 13110–13124. https://doi.org/10.1109/TWC.2024.3398770.
[19] M. Cui, L. Dai, Channel estimation for extremely large-scale MIMO: far-field or near-field?, IEEE Trans. Commun. 70 (4) (2022) 2663–2677. https://doi.org/10.1109/TCOMM.2022.3146400.
[20] W. Liu, C. Pan, H. Ren, J. Wang, R. Schober, Near-field multiuser beam-training for extremely large-scale MIMO systems, IEEE Trans. Commun. 73 (4) (2025) 2663–2679. https://doi.org/10.1109/TCOMM.2024.3468208.
[21] T. Zheng, M. Cui, Z. Wu, L. Dai, Near-field wideband beam training based on distance-dependent beam split, IEEE Trans. Wireless Commun. 24 (2) (2025) 1278–1292. https://doi.org/10.1109/TWC.2024.3507795.
[22] K. Huang, J. Guan, Y. Wang, X. Luo, A symmetric beam training method for near-field XL-MIMO systems, in: Proceedings of the 6th International Conference on Electrical Engineering and Information and Communication Engineering (EEICE), Shenzhen, China, 2025, pp. 1491–1496. https://doi.org/10.1109/EEICE65049.2025.11033820.
[23] C. Zhou, C. Wu, C. You, J. Zhou, S. Shi, Near-field beam training with sparse DFT codebook, IEEE Trans. Commun. 73 (6) (2025) 4394–4408. https://doi.org/10.1109/TCOMM.2024.3511702.
[24] J. Suh, C. Kim, W. Sung, J. So, S.W. Heo, Construction of a generalized DFT codebook using channel-adaptive parameters, IEEE Commun. Lett. 21 (1) (2017) 196–199. https://doi.org/10.1109/LCOMM.2016.2606432.
[25] W. Liu, Z. Wang, Statistics-assisted beam training for mmwave massive MIMO systems, IEEE Commun. Lett. 23 (8) (2019) 1401–1404. https://doi.org/10.1109/LCOMM.2019.2922632.
[26] C. Zhang, C. Qi, A. Nallanathan, Fast multibeam training for RIS-assisted millimeter wave massive MIMO, IEEE Commun. Lett. 28 (1) (2024) 168–172. https://doi.org/10.1109/LCOMM.2023.3333683.
[27] C. You, B. Zheng, R. Zhang, Fast beam training for IRS-assisted multiuser communications, IEEE Wireless Commun. Lett. 9 (11) (2020) 1845–1849. https://doi.org/10.1109/LWC.2020.3005980.
[28] H. Wang, J. Fang, H. Duan, H. Li, Fast hybrid far/near-field beam training for extremely large-scale millimeter wave/terahertz systems, IEEE Trans. Commun. 73 (6) (2025) 3956–3970. https://doi.org/10.1109/TCOMM.2024.3502681.
[29] J. Zhang, M. Li, M.-J. Zhao, X. Ji, W. Xu, Multi-user beam training and transmission design for covert millimeter-wave communication, IEEE Trans. Inf. Forensics Security 17 (2022) 1528–1543. https://doi.org/10.1109/TIFS.2022.3163583.
[30] P. Wang, J. Fang, W. Zhang, H. Li, Fast beam training and alignment for IRS-assisted millimeter wave/terahertz systems, IEEE Trans. Wireless Commun. 21 (4) (2022) 2710–2724. https://doi.org/10.1109/TWC.2021.3115152.
[31] F. Liu, C. Masouros, A tutorial on joint radar and communication transmission for vehicular networks-Part II: state of the art and challenges ahead, IEEE Commun. Lett. 25 (2) (2021) 327–331. https://doi.org/10.1109/LCOMM.2020.3025339.
[32] D. Zhu, J. Choi, R.W. Heath, Auxiliary beam pair enabled AoD and AoA estimation in closed-loop large-scale millimeter-wave MIMO systems, IEEE Trans. Wireless Commun. 16 (7) (2017) 4770–4785. https://doi.org/10.1109/TWC.2017.2702617.
[33] L. Zhu, J. Zhang, Z. Xiao, X. Cao, D.O. Wu, X.G. Xia, 3-D beamforming for flexible coverage in millimeter-wave UAV communications, IEEE Wireless Commun. Lett. 8 (3) (2019) 837–840. https://doi.org/10.1109/LWC.2019.2895597.
[34] Y. Chen, J. Tan, M. Hao, R. MacKenzie, L. Dai, Accurate beam training for RIS-assisted wideband terahertz communication, IEEE Trans. Commun. 71 (12) (2023) 7425–7440. https://doi.org/10.1109/TCOMM.2023.3317291.
[35] M. Cui, L. Dai, Near-field wideband channel estimation for extremely large-scale MIMO, Sci. China Inf. Sci. 66 (7) (2023) 172303. https://doi.org/10.1007/s11432-022-3654-y.
[36] M. Cui, L. Dai, Z. Wang, S. Zhou, N. Ge, Near-field rainbow: wideband beam training for XL-MIMO, IEEE Trans. Wireless Commun. 22 (6) (2023) 3899–3912. https://doi.org/10.1109/TWC.2022.3222198.
[37] C. Zhou, C. You, S. Shi, J. Zhou, C. Wu, Super-resolution wideband beam training for near-field communications with ultralow overhead, IEEE Internet Things J. 12 (14) (2025) 28793–28808. https://doi.org/10.1109/JIOT.2025.3568193.
[38] Y. Lu, Z. Zhang, L. Dai, Hierarchical beam training for extremely large-scale MIMO: from far-field to near-field, IEEE Trans. Commun. 72 (4) (2024) 2247–2259. https://doi.org/10.1109/TCOMM.2023.3344600.
[39] Y. Zhang, X. Wu, C. You, Fast near-field beam training for extremely large-scale array, IEEE Wireless Commun. Lett. 11 (12) (2022) 2625–2629. https://doi.org/10.1109/LWC.2022.3212344.
[40] X. Wei, L. Dai, Y. Zhao, G. Yu, X. Duan, Codebook design and beam training for extremely large-scale RIS: far-field or near-field?, China Commun. 19 (6) (2022) 193–204. https://doi.org/10.23919/JCC.2022.06.015.
[41] C. Wu, C. You, Y. Liu, L. Chen, S. Shi, Two-stage hierarchical beam training for near-field communications, IEEE Trans. Veh. Technol. 73 (2) (2024) 2032–2044. https://doi.org/10.1109/TVT.2023.3311868.
[42] J. Nie, Y. Cui, Z. Yang, W. Yuan, X. Jing, Near-field beam training for extremely large-scale MIMO based on deep learning, IEEE Trans. Mob. Comput. 24 (1) (2025) 352–362. https://doi.org/10.1109/TMC.2024.3462960.
[43] G. Jiang, C. Qi, Near-field beam training based on deep learning for extremely large-scale MIMO, IEEE Commun. Lett. 27 (8) (2023) 2063–2067. https://doi.org/10.1109/LCOMM.2023.3289513.